The below table summarizes the types of straight lines related to the given information.

S.No	Information given	Equation of the straight lines
1	Slope(m) and y-intercept (b)	$y = mx + b$
2	Slope (m) and point (x_1, y_1)	$y - y_1 = m(x - x_1)$
3	Two points(x_1, y_1) and (x_2, y_2)	$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$
4	x-intercept (a) and y-intercept(b)	$\frac{x}{a} + \frac{y}{b} = 1$
5	Normal length (p) , angle (α)	$x\cos\alpha + y\sin\alpha = p$
6	Parametric form: parameter-r	$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r$
7	The general equation	$ax + by + c = 0$

If we have two variable quantities, then each can be represented by a variable. If the rate of change of one variable with respect to the other variable is constant, then the relationship between them is linear.

In linear equation, the choice of one as independent and other as a dependent may represent the physical reality or may be convenient fiction. The independent variable is normally plotted on the horizontal axis (x-axis), the dependent variable on the vertical axis (y-axis). That is the values of x are always independent and y is dependent on those values of x.

The number scales on the two axes need not be the same. Indeed, in many applications different quantities are represented by x and y. For example, x may represent the number of mobile phones sold and y the total revenue resulting from the sales. In such cases it is often desirable to choose different number scales to represent the different quantities. However, that the zero of both number scales coincide at the origin of the two-dimensional coordinate system.

From the given information, to solve the problem using the concepts of straight lines, we have to select suitably one of the six equations given above.

Example 6.7 Find the slope of the straight line passing through the points (5,7) and (7,5). Also find the angle of inclination of the line with the x-axis.

Solution:

Let (x_1, y_1) and (x_2, y_2) be (5,7) and (7,5) respectively. Let θ be the angle of inclination of the line with the x-axis

Slope of the line $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5 - 7}{7 - 5} = -1$

We know that $m = \tan\theta$

That is, $\tan\theta = -1 \quad \Rightarrow \quad \theta = \frac{3\pi}{4}$ *or* 135°

Slope and angle of inclination of the line with the x-axis are respectively $m = -1$ and $\theta = \frac{3\pi}{4}$

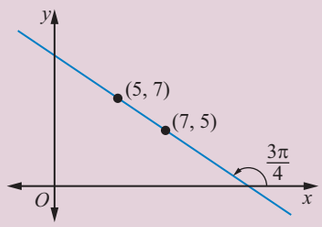


Figure 6.25